

Lecture 3 Live Problems

State variables and their change equations

Points toward lab manual pages: **BT1, BT2, Pop1, Pop2, SL1, SL2** (section L1 S2)

TO DO: No live problems existed for Lecture 3 in 2025. Candidates from the slides: (1) complete the cell-count table $t, \Delta t, C, \Delta C, \Delta C / \Delta t$ from real embryo data; (2) relative-change drills (200 cells $\pm 10\%$, $+100\%$, $+50\%$ twice); (3) build the caffeine change equation $\Delta C / \Delta t = 15 - 0.2C$ and iterate from $C = 0$. Alignment audit also asks for a semilog-axis warm-up here (SL1, SL2).

1. Synchronized cell division. Fill in the table for the growing embryo, and compute the rate of growth in each interval.

t (hours)	Δt	C (cells)	ΔC	$\Delta C / \Delta t$
0		1		
27		2		
40		4		
62		8		
90		16		
115		32		

2. Caffeine. Let C be the caffeine in my body (mg), t in hours. I ingest 15 mg per hour (on average) and metabolize 20% of what is in my body each hour. Write the change equation, then predict C over the first several hours starting from $C = 0$.